REQUIREMENT ANALYSIS AND IMPLEMENTATION PLAN

1. Requirement Analysis

1.1 Overview

The proposed Integrated Hotel Guest Request, Service Assignment, and Resolution Tracking System is a web-based application designed to improve how Swiss-Belhotel Blulane receives, processes, assigns, and monitors guest service requests.

Based on the hotel's consultation and interview feedback, the Front Desk will serve as the central coordinator of guest requests. The system will allow guests to submit service requests online, Front Desk personnel to review and forward accepted requests, department supervisors to assign staff, and service staff to update task progress.

The system will also maintain centralized records of requests, assignments, status changes, resolution notes, and timestamps to support monitoring and accountability.

1.2 Identification of System Users

The system will have four primary user roles.

User	Responsibilities
Hotel Guest	Submit service requests and view relevant request updates.
Front Desk Personnel	Review, accept, reject, mark unavailable, and route requests.
Department Supervisor	View requests, assign or reassign staff, and monitor progress.
Service Staff	View assigned tasks, update status, and record completion or unresolved notes.

Hotel management may also access summary reports if this access is approved by the hotel.

1.3 Functional Requirements

The functional requirements describe the specific operations and features that the system must perform.

ID	Functional Requirement	Description
FR-01	User Authentication	Allow authorized users to log in according to their assigned roles.

FR-02	Guest Request Submission	Allow guests to submit service requests through the web portal.
FR-03	Request Updates	Allow guests to view relevant updates regarding their submitted requests.
FR-04	Request Review	Allow Front Desk personnel to review incoming requests.
FR-05	Request Acceptance and Rejection	Allow Front Desk personnel to accept, reject, or mark requests as unavailable.
FR-06	Department Routing	Allow Front Desk personnel to forward accepted requests to the appropriate department.
FR-07	Staff Assignment	Allow department supervisors to assign suitable staff members to accepted requests.
FR-08	Staff Reassignment	Allow supervisors to reassign tasks when necessary.
FR-09	Task Monitoring	Allow supervisors to monitor the progress of assigned tasks.
FR-10	Status Updates	Allow service staff to update the progress of their assigned tasks.
FR-11	Resolution Notes	Allow service staff to record completed work or explain why a request remains unresolved.
FR-12	Request History	Maintain records of request details, assignments, reassignment, status changes, notes, and timestamps.
FR-13	Follow-up Monitoring	Allow Front Desk personnel and supervisors to monitor delayed and unresolved requests.
FR-14	Summary Reports	Generate reports based on request type, department, and status, including completed and unresolved requests.

1.4 Non-Functional Requirements

These requirements describe the expected quality and performance of the system.

ID	Requirement	Description
NFR-01	Usability	The system must have a simple and understandable interface for guests and hotel employees.
NFR-02	Security	The system must provide authentication and role-based access to protect request information.
NFR-03	Reliability	The system must properly record requests, assignments, and status updates.
NFR-04	Performance	The system should respond efficiently when users submit requests, access records, and update task statuses.
NFR-05	Data Integrity	The system must maintain accurate and consistent request records and status histories.
NFR-06	Accessibility	The system should be accessible through common web browsers on supported devices with an internet connection.
NFR-07	Maintainability	The system should have an organized structure that allows future modifications and improvements.

NFR-08	Compatibility	The system should function properly on supported desktop and mobile browsers.

1.5 Hardware Requirements

Hardware	Minimum Requirement
Development Computer	Intel Core i3 or equivalent
RAM	8 GB
Storage	256 GB SSD
Processor	Dual-core or better
Guest Device	Smartphone, tablet, or computer
Staff Device	Desktop computer, laptop, or tablet
Network	Stable internet or local network connection

1.6 Software Requirements

Software	Purpose
Windows 10/11	Development operating system
Visual Studio Code	Source code editor
PHP	Backend development
Laravel	Web application framework
MySQL	Database management
HTML	Web page structure
CSS	Interface styling
JavaScript	Client-side interaction
Bootstrap	Responsive interface design
XAMPP	Local development environment
Google Chrome	System testing and access

1.7 System Constraints and Limitations

The system will follow the limitations established in the research paper:

1. The system will focus only on hotel guest service requests.
2. Restaurant ordering and food service requests are excluded.
3. Online payments, invoices, and room charging are not included.
4. Real-time chat and live messaging are not included.
5. The system will not automatically assign staff members.
6. Front Desk personnel and supervisors will manually review and route requests.
7. The system will not include inventory management or stock-level monitoring.
8. Online reservation integration is excluded.
9. Check-in and check-out processing will remain outside the system.
10. Guest confirmation for every completed task will not be mandatory.

2. System Process Analysis

The proposed system will follow a centralized request-handling workflow.

2.1 Proposed System Workflow

1. **Hotel Guest** - Submits a service request through the web portal.

2. **Front Desk** - Reviews the request and accepts, rejects, or marks it unavailable.

3. **Front Desk** - Accepted requests are forwarded to the appropriate department.

4. **Department Supervisor** - Assigns or reassigns an appropriate service staff member.

5. **Service Staff** - Performs the task and updates its progress or resolution notes.

6. **Request Monitoring and Resolution** - Front Desk and supervisors monitor completed and unresolved requests.

2.2 Request Status Requirements

The system should maintain the following request statuses:

Status	Description
Pending Review	Request submitted and awaiting Front Desk review.
Accepted	Request approved by Front Desk.
Rejected	Request declined by Front Desk.
Unavailable	Requested item or service cannot be provided.
Assigned	Request forwarded and assigned to service staff.
In Progress	Service staff is currently handling the request.
Completed	Assigned staff has finished the task.
Unresolved	Task cannot be completed or requires further action.

Note: A task marked Completed by service staff does not automatically mean the guest's concern is fully resolved. Follow-up may still be necessary.

3. Implementation Plan

3.1 Implementation Approach

The researchers propose using an Agile development approach to develop the Integrated Hotel Guest Request, Service Assignment, and Resolution Tracking System.

This approach allows the development team to build the system in smaller stages, test individual features, gather feedback from intended users, and make necessary improvements before final implementation.

3.2 Development Phases

Phase	Activities	Expected Output
Phase 1: Requirement Gathering	Conduct interviews, identify user needs, and confirm hotel workflows.	Approved system requirements
Phase 2: System Analysis and Design	Prepare use case diagrams, process flowcharts, database design, and interface prototypes.	System design documentation
Phase 3: Database Development	Create tables for users, guests, departments, requests, assignments, and request history.	Functional database
Phase 4: Frontend Development	Develop the guest portal and staff dashboards.	User interfaces
Phase 5: Backend Development	Implement authentication, request processing, routing, assignment, and status updates.	Functional system modules
Phase 6: System Integration	Connect all modules and implement the complete request workflow.	Integrated web application
Phase 7: Testing and Evaluation	Perform functional testing, usability testing, reliability testing, and performance testing.	Testing and evaluation results
Phase 8: Deployment	Configure the system on the approved hosting or local network environment.	Deployed system
Phase 9: Maintenance	Correct errors and improve the system based on user feedback.	Updated and maintained system

3.3 Proposed Development Timeline

The following is an estimated 12-week implementation schedule.

Implementation schedule

Estimated duration of each development phase in weeks.

Week	Planned Activity
1–2	Requirement gathering and analysis
3–4	System design and prototyping
5	Database development
6–7	Frontend development
8–9	Backend development
10	System integration
11	Testing and evaluation
12	Deployment and documentation

Maintenance and improvements will continue after deployment.

3.4 Proposed Database Requirements

The database will store and organize the information needed for request management.

Database Table	Important Fields	Purpose
users	user_id, name, email, password, role	Stores system user accounts.
guests	guest_id, user_id, room_number	Stores guest information.
departments	department_id, department_name	Stores hotel departments.
requests	request_id, guest_id, request_type, description, status, created_at	Stores guest service requests.
assignments	assignment_id, request_id, supervisor_id, staff_id, assigned_at	Records staff assignments and reassignments.
request_history	history_id, request_id, previous_status, new_status, notes, updated_at	Maintains request activity history.
notifications	notification_id, user_id, message, is_read, created_at	Stores relevant system notifications.

The final database structure will be adjusted based on the confirmed hotel requirements.

3.5 System Testing and Evaluation Plan

The researchers will conduct testing to determine whether the developed system meets its intended requirements.

Testing Type	Purpose
Functional Testing	Verify that each system feature works according to its requirements.
Usability Testing	Determine whether users can easily navigate and operate the system.
Reliability Testing	Check whether the system consistently records requests and updates.
Performance Testing	Evaluate system response when submitting and retrieving requests.
Integration Testing	Verify that all modules work together correctly.
User Acceptance Testing	Gather feedback from intended users regarding system acceptability.

The evaluation will focus on four major criteria:

- Functionality: Does the system perform the required operations?
- Usability: Is the system easy to understand and operate?
- Reliability: Does the system maintain accurate and consistent records?
- Performance: Does the system respond efficiently to user actions?

4. Expected Implementation Output

After completing the development and implementation process, the researchers expect to produce a web-based system with the following components:

Guest Engagement and Operations Platform for Hotels | WeBee

Guest Portal

Guests can submit service requests and view relevant request updates.

Guest Experience Platform for Hotels and Resorts | STAY

Front Desk Dashboard

Provides a centralized inbox for reviewing and routing guest requests.

Habot Connect Task Management Dashboard (Design & Specification) by Shubham Singhania on Dribbble

Supervisor Dashboard

Allows supervisors to assign staff, reassign tasks, and monitor progress.

20 Best Practices & Examples for Better Dashboard Designs

Service Staff Dashboard

Displays assigned tasks and provides options for status updates and resolution notes.

Atlassian Analytics | Atlassian

Reports and Monitoring

Displays request summaries, department activity, completed tasks, and unresolved requests